

Difference Between JPA, Hibernate and Spring Data JPA

Spring Boot Notes

1 Introduction

JPA (Java Persistence API) is a specification used for mapping Java objects to relational databases.

Hibernate is one of the most popular ORM frameworks that implements the JPA specification.

Spring Data JPA is a Spring module that reduces boilerplate code by providing repository interfaces and automatic CRUD methods.

2 Comparison Table

Feature	JPA	Hibernate	Spring Data JPA
Definition	Java Persistence API. It is a specification for object relational mapping.	ORM framework that implements JPA.	Spring framework module built on top of JPA.
Type	Specification	Framework	Spring Library
Purpose	Defines persistence rules for Java applications.	Maps Java objects to database tables.	Simplifies CRUD operations using repositories.
Main Interface	EntityManager	Session	JpaRepository
Transaction Management	EntityTransaction	Transaction	@Transactional Annotation
Query Language	JPQL	HQL and JPQL	Derived Queries, JPQL
Configuration	Needs implementation provider.	Requires Hibernate configuration.	Very little configuration.
Boilerplate Code	High	Medium	Very Low
Advantages	Standard API, portable.	Caching, Lazy Loading, HQL support.	Automatic CRUD, Pagination, Sorting, Less Code.
Disadvantages	Needs implementation.	More configuration required.	Depends on Hibernate/JPA provider.

3 Code Examples

1. JPA Example

```
EntityManager em = emf.createEntityManager();

em.getTransaction().begin();

em.persist(employee);

em.getTransaction().commit();

em.close();
```

2. Hibernate Example

```
Session session = sessionFactory.openSession();

Transaction tx = session.beginTransaction();

session.save(employee);

tx.commit();

session.close();
```

3. Spring Data JPA Repository

```
public interface EmployeeRepository
extends JpaRepository<Employee, Integer> {

}
```

4. Spring Data JPA Service

```
@Service
public class EmployeeService {

    @Autowired
    private EmployeeRepository repository;

    @Transactional
    public void addEmployee(Employee employee){
        repository.save(employee);
    }
}
```

```
}  
}
```

4 Advantages

JPA

- Standard Java Persistence Specification
- Database Independent
- Easy to switch providers
- Portable API

Hibernate

- Implements JPA
- Supports Caching
- Supports Lazy Loading
- Uses HQL
- Better Performance

Spring Data JPA

- Automatic CRUD Operations
- Repository Pattern
- Query Method Generation
- Pagination and Sorting
- Minimal Boilerplate Code
- Easy Spring Boot Integration

5 When to Use

Requirement	Recommended Technology
Standard Persistence API	JPA
Advanced ORM Features	Hibernate
Spring Boot Projects	Spring Data JPA
Less Coding	Spring Data JPA
Complex Database Mapping	Hibernate

Conclusion

JPA defines the standard rules for persistence.

Hibernate implements JPA and provides ORM functionality.

Spring Data JPA builds on Hibernate and JPA to simplify database operations with repository interfaces, automatic CRUD methods, pagination, sorting, and transaction management.